MIL-C-60317C(PA)
15 April 1974
SUPERSEDING
MIL-C-60317B(MU)
19 December 1968

MILITARY SPECIFICATION

CARTRIDGE, 60MM, SMOKE, WP, M302A1 ASSEMBLING, MARKING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the assembling, marking and packing for one type of smoke cartridge designated as M302A1, for use with 60MM Mortars.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-Ammunition and Special Weapons; General Specification for
MIL-P-60316	-Primer, Percussion, M32, Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-1168	-Lot Numbering of Ammunition
MIL-STD-1235	-Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

PRODUCT AND PACKING DRAWINGS

9215576	-Container, Ammunition, Fiber M567 for Cartridge, 60MM Smoke, WP, M302A1 in Jungle Wrap

FSC: 1310

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9215577 -Box, Ammo, Packing, Wood, Nailed for Cartridge, 60MM Smoke, WP, M302A1, in Jungle Wrap
9215575 -Cartridge, 60MM, Smoke, WP, M302A1

INSPECTION EQUIPMENT DRAWINGS

7256281 -Alignment Check Gage
7256348 -Alignment Gage
7257962 -Limit Check Gage

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal, shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 0-190 -Code of Federal Regulations Title 49 (Revised as of January 1 1968).

The Department of Transportation Regulations are now a part of the Code of Federal Regulations (1968 Edition and revisions) available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite "49 CFR 0-190 (Revised as of 1 January 1968).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Cartridges.-The cartridge shall comply with all requirements specified on Drawing (dwg.) 9215575 and with all requirements specified in applicable specifications.

3.3 Proving ground test.-The cartridge shall comply with the following requirements when tested as specified in 4.3.2.

3.3.1 Failure to fire.-The cartridge shall not fail to fire.

3.3.2 Safety.-There shall be no premature functioning in the mortar tube or in flight before impact.

3.3.3 Impact functioning.-There shall be visible evidence of functioning upon impact as indicated by presence of a flash or smoke cloud (see 6.5).

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3.3.4 Horizontal range

3.3.4.1 No cartridge shall have a horizontal range less than 80 percent of the average horizontal range.

3.3.4.2 The standard deviation in horizontal range shall be no greater than the maximum (max.) lot values shown in Table I for the increment charge indicated.

TABLE I

Increment Charge Zone	Max. Lot Standard Deviation (Meters)
0	12
4	32

3.4 Workmanship

3.4.1 All processing and assembly operations of the cartridge shall be correctly performed in a thorough workmanlike manner. Parts shall be free of chips, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.4.2 The painting and marking shall be restored by the assembling contractor in any component furnished to him painted and marked when such painting and marking has been marred or obliterated.

3.5 Initial Production Test.-The cartridges shall function satisfactorily in accordance with the requirements of 3.3 after being subjected to the sequence of environmental tests in 4.2.3.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the

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Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded complete round cartridges shall contain:

a. WP loaded body filling assemblies from one manufacturer in accordance with the same drawing, same drawing revision, same specification and same specification revision of one loading lot interfix number.

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b. Loaded ignition cartridges of one lot interfix number from one manufacturer.
c. Fin assemblies of one lot interfix number from one manufacturer.
d. Loaded primers of one lot interfix number from one manufacturer.
e. Propellant increments made from not more than one lot of propellant from one manufacturer.
f. Loaded fuzes of one lot interfix number from one manufacturer.
g. Loaded projectiles bursters of one lot interfix number from one manufacturer.
h. Fin extension of one lot interfix number from one manufactuer.

4.2.2 Examination.-The examination listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235, may be used if approved by the Government. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Body filling assembly as received from WP loading facility (see dwg. 9205339 covering a detail of dwg. 9215575).

Categories	Defects	Method Of Inspection
Critical:		
	1. WP leakage or evidence of leakage	Visual
Major:	None defined	
Minor:		
	201. Evidence of poor workmanship (see 3.4)	Visual

4.2.2.2 Cartridge, prior to inserting burster (see dwg. 9215575).

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Categories	Defects	Method Of Inspection
Critical:		
1.	Burster pad missing	Visual
Major:		
101.	Foreign matter in burster cavity	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.4)	Visual

4.2.2.3 Cartridge, prior to assembling fuze (see dwg. 9215575).

Categories	Defects	Method Of Inspection
Critical: None defined.		
Major:		
101.	Burster charge missing	Visual
102.	Foreign matter in fuze well	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.4)	Visual

4.2.2.4 Cartridge, prior to assembling primer (see dwg. 9215575).

Categories	Defects	Method Of Inspection
Critical:		
1.	Ignition cartridge missing or inverted	Visual
Major:		
101.	Fin or fin extension loose (see 4.3.1)	Test
102.	Fuze loose (see 4.3.1)	Test
103.	Fin extension to fin and shell body sealing compound missing	Visual
Major:		
104.	Sealing compound bead improperly located, missing or not full 360 degrees in base of cartridge housing cavity (Prior to insertion of ignition cartridge)	Visual

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105. Sealing compound bead improperly located, missing or not full 360 degrees on red end of ignition cartridge (Prior to seating of primer into fin cavity).....Visual
106. Excessive sealing compound on primer pellet closing disc.....Visual

Minor:
201. Foreign matter in cavity.....Visual

4.2.2.5 Cartridge (see dwg. 9215575)

Categories	Defects	Method Of Inspection
Critical:		
1.	Primer head above flush with rear surface of fin cartridge container	Straight-Edge
2.	Fin damaged or distorted (this characteristic must be gaged after all mechanical handling of the fin)	Gage (CD)
3.	Fuze safety wire missing or insecure	Visual-Manual
Major:		
101.	Total length, max	Gage (CD)
102.	Primer stake missing or thread sealing compound does not cover a minimum of one half the number of male primer threads for a full 360 degrees (after disassembly) (as applicable)	Visual
103.	Any component damaged or incorrectly assembled (except as described by Critical defect 2 above)	Visual
104.	Primer fails to withstand reapplication of minimum assembly torque	Test
Minor:		
201.	Protective coating with total damaged area in excess of 1/4 square inch	Visual
202.	Marking misleading, missing or unidentifiable, including incorrect color code	Visual
203.	Fuze not fully seated	Visual
204.	Evidence of poor workmanship (see 3.4)	Visual

4.2.2.6 Unsealed fiber container (see dwg. 9215576).

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Categories	Defects	Method Of Inspection
Critical:		
	1. Propellant increment missing	Visual
Major:	100 percent inspection	
	101. Propellant increment cut or torn	Visual
Major:		
	102. Ring support missing	Visual
	103. Filler assembly missing	Visual
Minor:		
	201. Glue or asphalt on round	Visual
	202. Cartridge improperly assembled into container	Visual
	203. Round cannot be removed by hand	Manual

4.2.2.7 Sealed fiber container (see dwg. 9215576).

Categories	Defects	Method Of Inspection
Critical: None defined.		
Major:		
	101. Container damaged or cut through all asphalt impregnated layers	Visual
	102. Tape incomplete, badly wrinkled or missing	Visual
	103. Metal end loose or distorted	Visual-Manual
Minor:		
	201. Sealing strip improperly applied	Visual
	202. Gap between cover and body (excessive (exceeds 1/8 inch)	Visual
	203. Contents loose	Manual
	204. Marking	Visual

4.2.2.8 Fiber container wrapped in barrier material prior to waxing - Jungle Wrap (see dwg. 9215576).

Categories	Defects	Method Of Inspection
Critical: None defined.		
Major:		
	101. Barrier material improperly assembled	Visual
	102. Barrier material ends improperly folded or taped	Visual

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Minor:
201. Marking missing or incomplete.....Visual

4.2.2.9 Fiber container after waxing - Jungle Wrap (see dwg. 9215576).

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:		
	101. Wax coating incomplete or damaged.....Visual	
Minor:		
	201. Marking missing or unidentifiable.Visual	
	202. Kraft paper improperly placed on container.....Visual	
	203. Tear tab not visible.....Visual	

4.2.2.10 Sealed wood packing box (see dwg. 9215577)

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:		
	101. Contents exposed.....Visual	
	102. DOD symbol missing or incorrect...Visual	
	103. Board broken or split.....Visual	
	104. Hardware or strapping missing, broken or loose.....Visual-Manual	
Minor:		
	201. Hardware improperly engaged.....Visual-Manual	
	202. Car seal missing, unsealed or improperly positioned.....Visual-Manual	
	203. Contents loose.....Visual-Manual	
	204. Handle missing or insecure.....Visual-Manual	
	205. Marking misleading, missing or unidentifiable.....Visual	

4.2.3 Testing

4.2.3.1 Check test for possible deterioration of Primer, Percussion, M32 (see Specification MIL-P-60316). If the total elasped time between original acceptance of the M32 primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the M32 primers have been

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subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in Specification MIL-P-60316 before the M32 primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges (see 6.4).

4.2.3.2 Profile and alignment (see dwg. 9215575).-Major defect-The cartridges, including those which have had the bourrelet repainted (see 3.4.2), shall be gaged 100 percent for profile and alignment. Any cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 2.1 and 4.2.4.

4.2.3.3 Proving ground test.-The cartridges shall be observed for the characteristics specified in Table II.

TABLE II

		Defect Classification	Accept	Reject
(a)	Failure to fire (see 3.3.1)	Major	1	2
(b)	Safety (see 3.3.2)	Critical	0	1
(c)	Impact functioning (see 3.3.3)	Major	3	4
(d)	Horizontal range (see 3.3.4.1) less than 80 percent of average horizontal range but not less than 60 percent of average horizontal range	Major	1	2
(e)	Horizontal range less than 60 percent of average horizontal range.	Critical	0	1
(f)	Range standard deviation (see 3.3.4.2)	Major		

4.2.3.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, a sample of 80 cartridges shall be selected from each lot for this test. Acceptance criteria shall be in accordance with Table II, and in addition, if the standard deviation in horizontal range for the sample cartridges fired at charge zone zero or charge zone four multiplied by the factor 0.79 exceeds the applicable maximum limit specified in Table I (see 3.3.4.2), the lot shall be rejected. The test shall be performed as specified in 4.3.2.

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4.2.3.3.2 After three consecutive lots have met the criteria of 4.2.3.3.1, a sample of 50 cartridges shall be selected from each lot for this test. Acceptance criteria shall be in accordance with Table II and in addition, if the standard deviation in horizontal range for the sample cartridges fired at charge zone zero or charge zone four multipled by the factor 0.75 exceeds the applicable maximum limit specified in Table I (see 3.3.4.2), the lot shall be rejected. The test shall be performed as specified in 4.3.2.

4.2.4 The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.2.2 and 4.2.3 (see 6.3).

4.3 Test methods and procedures.

4.3.1 Torque test.-The assembly torque shall be applied to the applicable component while holding the adjacent mating component in such a manner as to avoid possible damage to any component of the assembly.

4.3.2 Proving ground test.-The test shall be performed at a Government proving ground in accordance with the applicable Acceptance Test Procedures. The cartridge shall be fired from a 60MM Mortar for which the round is standard. One half of the sample cartridges shall be fired with an ignition cartridge (Charge zone zero) and one half of the sample cartridges shall be fired with an ignition cartridge and four propellant increments (charge zone four). The individual velocities, max pressures and impact ranges shall be obtained and recorded. The standard deviation and the arithmetic average of the impact ranges shall be calculated and recorded.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The cartridges shall be preserved and packaged in accordance with dwg. 9215576.

5.2 Packing

5.2.1 Level A.-The cartridges in fiber containers shall be packed in accordance with dwg. 9215577.

5.3 Marking.-Marking of the wood box shall be in accordane with dwg. 9215577. The fiber container shall be marked in accordance with dwg. 9215576.

6. NOTES

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6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Equipment Designs.-Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs, if available, are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.2.2 and 6.2.3 should he elect to do so.

6.2.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which established the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.2.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposed to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial, or with SIE designs if not otherwise covered thereby (see 6.2.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number, paragraph

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number and defect number if such number has been assigned shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.2.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.3 Cost of check test.-The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.4 Impact functioning.-The cartridge shall function with sufficient fragmentation of the projectile to adequately disperse the chemical.

6.5 Submission of test data.-In addition to the normal distribution of records, when the cartridge is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to the following:

Commander
Picatinny Arsenal
ATTN: SARPA-QA
Dover, New Jersey 07801

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1310-A-056

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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